

Presentation Outline

March 20, 2026

1. Title
2. Motivation
3. Ionization vs displacement damage
4. Geant4 model
5. Geant4 outputs
6. Defect model
7. Lifetime model
8. Device degradation metrics
9. Summary

Cryogenic extension

Add a short ending section on how radiation-induced defect behavior changes at low temperature. Focus on migration freeze-out, trap emission freeze-out, and annealing freeze-out, and use simple MATLAB Arrhenius plots to show the trend from 300 K toward the cryogenic limit.